VERA: Support-Aware Shift Envelopes for Representation Erasure

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Abstract

Concept erasers can suppress sensitive information in a representation while damaging its target task or failing under deployment shift. The eraser alone does not answer the deployment question: under what shift, if any, is an edit still supported by evidence? We introduce VERA, Verified Erasure under Reweighting Ambiguity, a decision layer that audits target loss on paired identity/edit outcomes and sensitive leakage with fresh heterogeneous attackers. For each registered edit, VERA reports a simultaneous support-aware envelope of environment and source-conditional density-ratio budgets under which all target-harm and balanced-leakage contracts hold, together with a more powerful fixed-profile decision, or returns ABSTAIN. We prove uniform envelope validity, permit arbitrary mixtures of observed environments and arbitrary source-prior shift, and show that no validation-only protocol can nontrivially certify unconstrained deployment mass outside its support. Finite-family testing and abstention are inherited from Learn Then Test style risk control, not claimed as new. The empirical protocol combines a 54-cell exact abstention study, a 216-cell candidate-family/group-count coverage grid, and a preregistered 200-run official-code matrix spanning five erasers, five datasets, and eight untouched seeds. VERA gives finite-sample false-acceptance control over its declared shift class and identifies when certification is impossible; on Camelyon17 it forced abstention in all 144 registered VERA configurations because the deployment hospital was outside certification support.Across 32 prespecified stress configurations, validation-only selection deployed contract-violating edits in 21.9% versus 0.0% for VERA. Across all threshold configurations, VERA retained 30.8% of external-oracle opportunities; the seed-blocked comparison did not survive Holm correction.

1 Introduction

Representation editing is an intervention. It changes what downstream models can recover from a frozen representation, often to remove a protected, nuisance, or source concept. INLP, R-LACE, LEACE, kernel methods, TaCo, SPLINCE, and MANCE++ provide increasingly expressive ways to construct such edits (Ravfogel et al. 2020, 2022a; Belrose et al. 2023; Ravfogel et al. 2022b; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Their successes do not make every resulting edit deployable. Removing a correlated concept can damage the target, a probe used by the eraser can understate leakage to a different attacker, and an IID validation result need not survive reweighting at deployment.

This paper asks a narrower, operational question: what is the largest declared shift under which one candidate edit can be certified to preserve the target while limiting all registered leakage attackers? The target audit is paired. For the same example, it subtracts the identity representation's zero-one loss from the edited representation's loss, isolating incremental harm rather than conflating it with baseline error. The shift class is stratified: within each environment, the target law has a bounded density ratio relative to its certification law. Leakage is the equal mean of two robust source-class recalls, each under a bounded source-conditional density ratio. Environment weights and source prevalence may therefore vary arbitrarily; ordinary prevalence-weighted accuracy is not the contract. At a declared profile, VERA uses a candidate-wise intersection–union test. Separately, it simultaneously upper-bounds every candidate-contract curve and inverts them into a support-aware shift envelope and common radius $\widehat{\Gamma}_e$. A zero radius means that even the IID contract is not established. New support forces abstention.

The statistical mechanism must be positioned carefully. Risk-controlling prediction sets, Learn Then Test (LTT), Conformal Risk Control, and Pareto Testing already calibrate finite algorithm families against one or several risks; recent work also couples risk, acceptance, and utility and provides group-conditional PAC control (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023; Yu and Liu 2026; Huang et al. 2026). Prompt Risk Control applies related candidate-wise bounds to LLM prompts and includes an estimated covariate-shift correction (Zollo et al. 2024). Weighted risk control and robust validation already address important forms of covariate and distribution shift (Tibshirani et al. 2019; Almeida et al. 2025; Cauchois et al. 2024; Ai and Ren 2024). VERA uses these foundations. Its proposed contribution is the representation-intervention object they do not report: a simultaneous, support-aware envelope of groupwise reweighting budgets for paired edit harm and an entire retrained-attacker portfolio. The common shift radius is one summary of that envelope; unsupported required cells mark the boundary at which it cannot be identified.

Our contributions are threefold. We robustify balanced accuracy into a source-prior-invariant leakage contract and certify its joint support-aware envelope with environment-conditional paired harm over a continuum of deployment budgets. We formalize unsupported-environment abstention by an indistinguishability theorem rather than a heuristic. We then test whether certification changes deployment decisions in a locked cross-modal study with official INLP, R-LACE, LEACE, TaCo, and MANCE++ code. All numbers are regenerated from per-run receipts; internal audits are not substitutes for scientific review.

2 Related Work

Concept erasure and probing. Linear erasure includes iterative nullspace projection, adversarial low-rank projection, and the closed-form LEACE solution (Ravfogel et al. 2020, 2022a; Belrose et al. 2023). Kernel erasure, rate-distortion objectives, unaligned-attribute removal, TaCo, SPLINCE, and MANCE extend the construction space (Ravfogel et al. 2022b; Basu Roy Chowdhury et al. 2023; Shao, Ziser, and Cohen 2023; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Nonlinear guardedness can remain after linear removal, while perfect erasure faces fundamental utility limits (Akbari, Afshari, and Boddeti 2025; Chowdhury et al. 2025). FARE and MMD-B-Fair provide important certificates for restricted fair-representation objectives (Jovanovic et al. 2023; Deka and Sutherland 2023). VERA neither replaces these erasers nor certifies absence of all information. It audits their candidate outputs against a fixed portfolio. That portfolio follows the broader lesson that a probe's result depends on its hypothesis class and training protocol (Hewitt and Liang 2019; Pimentel et al. 2020; Voita and Titov 2020; Elazar et al. 2021).

Risk control and abstention. LTT turns calibration into testing over a finite family, and Pareto Testing extends the view to multiple risks and objectives (Angelopoulos et al. 2025; Laufer-Goldshtein et al. 2023). Conformal Risk Control treats general monotone losses (Angelopoulos et al. 2024). Joint adaptive certificates couple risk, acceptance, and utility, while group-conditional PAC methods control routing risk within registered groups (Yu and Liu 2026; Huang et al. 2026). These works invalidate any claim that VERA invents finite-family, joint, or groupwise acceptance. Prompt Risk Control is a particularly close application template: it selects LLM prompts using high-probability bounds on mean, tail, and dispersion risks and corrects covariate shift using unlabeled target data (Zollo et al. 2024). VERA instead audits paired representation interventions without a target sample or estimated importance weights, reports worst-case guarantees over a declared bounded reweighting class, retrains a heterogeneous attacker portfolio, and makes unsupported cells an explicit impossibility boundary. Selective classification and SelectiveNet reject individual examples (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017, 2019); VERA instead rejects an entire intervention. Continuous monitoring detects risk violations after deployment (Timans et al. 2025), whereas VERA is a predeployment, support-scoped sensitivity certificate.

Distribution shift and robust evaluation. Known importance weights can restore conformal or high-probability risk control under a specified covariate shift (Tibshirani et al. 2019; Almeida et al. 2025). Robust validation and fine-grained robust conformal inference provide ambiguity-set guarantees (Cauchois et al. 2024; Ai and Ren 2024), while robust model evaluation and CVaR concentration supply closely related technical tools (Najafi, Sani, and Farnia 2025; Thomas and Learned-Miller 2019; Waudby-Smith and Ramdas 2024). Distributional fairness certificates and Wasserstein fairness audits are close application-level neighbors (Kang et al. 2022; Ehyaei, Farnadi, and Samadi 2025). Worst-case subpopulation treatment effects are especially close to our paired-harm view (Jeong and Namkoong 2020). VERA does not claim these tools or generic robust fairness auditing. It applies bounded reweighting to a selected representation edit's paired target and retrained-attacker contracts and reports the admissible support-scoped envelope. Group DRO and robustness benchmarks motivate the observed-group view (Sagawa et al. 2020; Koh et al. 2021; Sohoni et al. 2020), and domain-adaptation impossibility results motivate explicit support assumptions (David et al. 2010; Subbaswamy, Adams, and Saria 2021).

3 Problem Setup

Let $Z \in \mathbb{R}^d$ be a frozen representation, Y its target label, and S a source or sensitive label. An edit $e \in \mathcal{E}$ is fitted without the certification fold and maps Z to Z^e. The identity is $e = 0$. Fresh target models are fitted after every edit. For certification example i, define paired target harm

$$H_{e,i} = \mathbf{1}\{f_e(Z_i^e) \neq Y_i\} - \mathbf{1}\{f_0(Z_i) \neq Y_i\} \in \{-1, 0, 1\}. \tag{1}$$

In the real protocol, identity is the paired comparator rather than a selectable erasure action. If a deployment protocol may select identity, it must register identity as one additional candidate in the finite-family correction; the separate family-size coverage grid does so. For each preregistered attacker $a \in \mathcal{A}$, let $L_{e,a,i} = \mathbf{1}\{a_e(Z_i^e) = S_i\} \in \{0, 1\}$. Attackers are retrained after each edit using construction data, but are fixed before certification outcomes are used.

For validation law P and $\Gamma \geq 1$, define

$$\begin{aligned} \mathcal{Q}_\Gamma(P) &= \left\{ Q \ll P : 0 \leq \frac{\mathrm{d}Q}{\mathrm{d}P} \leq \Gamma \quad P\text{-a.s.} \right\}, \\ \rho_{\Gamma,P}(V) &= \sup_{Q \in \mathcal{Q}_\Gamma(P)} \mathbb{E}_Q[V]. \end{aligned} \tag{2}$$

The target contract uses P_g, the certification law conditional on environment g. For binary source, leakage uses the balanced robust accuracy

$$B_{e,a}(\eta_0, \eta_1) = \frac{1}{2} \sum_{s \in \{0,1\}} \rho_{\eta_s, P_s}(L_{e,a}), \tag{3}$$

where P_s is source-conditional. Balanced accuracy itself is standard (Brodersen et al. 2010); our change is to robustify

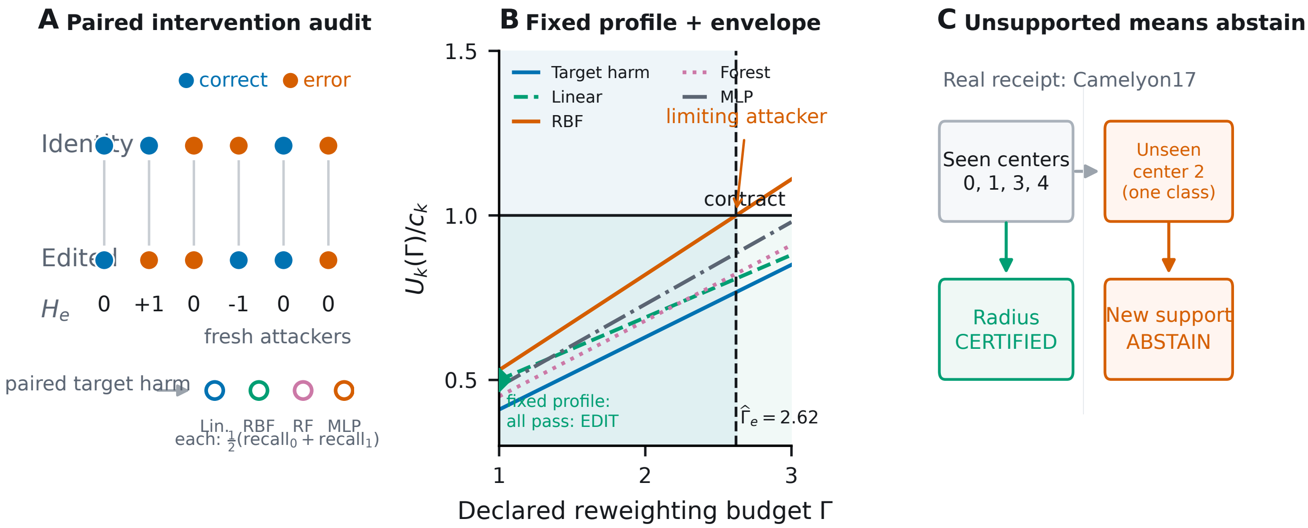

Figure 1: **VERA certifies an edit only within a declared, supported shift class.** (A) Target harm compares identity and edited predictions on the same examples; fresh heterogeneous attackers measure equal-weight source-class recall. (B) At a fixed profile, every component must pass the candidate IUT; simultaneous upper-risk curves are separately inverted to obtain the shift envelope and limiting common radius. Curves are schematic. (C) The guarantee cannot cross new support. In the registered Camelyon17 split, certification observes centers 0, 1, 3, and 4, while external center 2 is absent, so a five-center deployment claim requires abstention unless an additional cross-center assumption is supplied.

each recall and couple the resulting attacker contracts to paired edit harm. A constant attacker scores $1/2$, not one, and changing source prevalence does not change the estimand. At zero or unit prevalence this is a standardized two-conditional functional, not ordinary accuracy observable in an absent class. The population common erasure shift radius is

$$\Gamma_e^\star = \sup\{\Gamma \ge 1 : \rho_{\Gamma,P_k}(V_{e,k}) \le c_k \quad \text{for every registered contract } k\}, \tag{4}$$

with value zero if a contract fails at $\Gamma = 1$.

This radius is a sensitivity statement, not an estimate of how much shift will occur. A user declares a profile of plausible density-ratio caps; VERA states whether the available certification sample supports the contracts throughout that profile. Conditioning target harm on environment prevents a favorable environment mixture from hiding damage in a smaller group. Standardizing leakage to equal source-class weight similarly prevents source prevalence from making an attacker appear weak. These two conditionings need not describe one factorized joint model: the claim applies when each registered conditional law obeys its own stated cap.

4 VERA

The bounded-reweighting risk has the CVaR dual

$$\rho_{\Gamma,P}(V) = \inf_{\eta \in [a,b]} \{\eta + \Gamma \mathbb{E}_P[(V-\eta)_+]\}. \tag{5}$$

For general bounded variables, a Dvoretzky–Kiefer–Wolfowitz band gives a uniform upper curve. Our discrete audit variables admit tighter exact curves. For Bernoulli leakage with success probability p, $\rho_{\Gamma,P}(L) = \min(1, \Gamma p)$. For paired harm with probabilities (p_-, p_0, p_+) on $(-1, 0, 1)$, set $a = \Gamma p_+$ and $b = \Gamma(1 - p_-)$. The adversary places its permitted mass on $+1$, then zero, then -1:

$$\rho_{\Gamma,P}(H) = \begin{cases} 1, & a \ge 1, \\ a, & a < 1 \le b, \\ a + b - 1, & b < 1. \end{cases} \tag{6}$$

VERA substitutes one-sided Clopper–Pearson bounds. For paired harm it uses an upper bound U_+ for p_+ and lower bound L_- for p_-, clips $\widetilde{p}_+ = \min(U_+, 1 - L_-)$ to preserve the simplex, and splits the component error between the two tails. Balanced leakage averages two class recall bounds, splitting its component error between the source classes.

At the fixed deployment profile, each candidate is an intersection–union test: all of its target and balanced-leakage components must pass at level $\delta/|\mathcal{E}|$. If a candidate is unsafe, false acceptance implies undercoverage of at least one fixed violated component. A union bound is needed over candidates, but not over components. The separately reported envelope uses Bonferroni coverage over every candidate–contract pair because every curve remains available for post-certification inspection.

Let $U^T_{e,g}(\gamma_g)$ denote the simultaneous target-harm upper curve in observed environment g, and let $U^L_{e,a,s}(\eta_s)$ denote attacker a's source-class recall curve. VERA's central reported object is

$$\widehat{\mathcal{E}}_e = \{(\boldsymbol{\gamma}, \boldsymbol{\eta}) : U^T_{e,g}(\gamma_g) \le \tau \quad \forall g, \quad \tfrac{1}{2}\textstyle\sum_s U^L_{e,a,s}(\eta_s) \le \lambda \quad \forall a\}. \tag{7}$$

Coordinates for unsupported required cells are zero. The

Algorithm 1 VERA fixed-profile selection and shift envelope

Require: Registered edits $\mathcal{E}$, contracts c_k, error δ, budget cap $\Gamma_{\max}$

1: Fit every edit, target model, and leakage attacker without certification outcomes
2: Allocate $\delta/|\mathcal{E}|$ to each candidate's fixed-profile IUT
3: **for** each edit e **do**
4: Accept the profile only if every component bound clears its threshold
5: Separately construct simultaneous curves and envelope $\widehat{\mathcal{E}}_e$
6: **end for**
7: **return** the registered utility-best accepted edit, its envelope, or ABSTAIN

common radius is the largest diagonal point $(\Gamma\mathbf{1}, \Gamma, \Gamma)$ in this envelope, right-censored at the registered computational cap. Reporting the vector envelope as well as its diagonal summary reveals which environment, source class, or attacker limits an edit.

Theorem 1 (Fixed-profile VERA)**.** *Fix M candidate edits and all audit variables independently of certification. At one declared shift profile, accept a candidate only if every component upper bound clears its threshold at level δ/M. Then the probability of deploying any candidate that violates at least one declared target or balanced leakage contract is at most δ.*

For an unsafe candidate, fix one violated population component. Accepting that candidate implies undercoverage of this component; intersecting more tests cannot increase that probability. Union bounding only over the M candidates proves the result. This is the LTT-style finite-family decision used for the primary experiment.

The distinction from simply applying LTT is in the population contract being tested. Each edit induces paired target variables and a family of retrained attacker variables; each variable is robustified over a continuum of conditional reweightings; and all coordinates are inverted into an admissible shift region. LTT supplies the fixed-family testing logic. The robust edit-specific envelope, balanced conditional leakage functional, and explicit support boundary define the deployment object studied here.

The simultaneous envelope is richer but more conservative. It contains every profile $(\boldsymbol{\gamma}, \boldsymbol{\eta})$ for which each environment target curve clears τ and each attacker's mean of two source-class curves clears λ. Its coverage event is uniform over the continuum of profile coordinates. Environment mixing preserves the target threshold by linearity, and source prevalence does not enter balanced leakage. Restricting this envelope to the diagonal gives the reported common radius. Full statements and proofs are in the supplement.

Theorem 2 (Support-aware envelope)**.** *With probability at least $1-\delta$, simultaneously for every registered edit e, every profile in $\widehat{\mathcal{E}}_e$ satisfies all target-harm and balanced-leakage contracts. Environment mixture weights and source prevalence may be selected after certification without another multiplicity correction.*

The proof event covers every edit, environment, attacker, source class, and every $\Gamma \geq 1$. Conditional robust-risk bounds imply each component contract; arbitrary mixtures of controlled environment risks remain controlled by linearity, and source prevalence is absent from the standardized leakage functional. Intersecting or choosing objects already covered by the same event does not spend additional error probability. This argument establishes the external-distribution guarantee only for laws inside the declared envelope.

Theorem 3 (Unsupported-cell impossibility)**.** *Suppose a required audit cell has zero validation probability, and the model class contains safe and unsafe worlds that induce identical protocol observations but disagree on the contract in that cell. If deployment may concentrate there, any validation-only protocol with uniform false-acceptance control δ accepts the edit in the safe world with probability at most δ.*

The proof is a two-world indistinguishability argument. It applies to an unseen environment or a missing environment–source-label cell; a single-class slice cannot identify the omitted class's conditional leakage. Camelyon17 center 2 and its constant binary source label therefore trigger abstention. This does not assert that the measured center-2 target outcome necessarily violates a contract.

The impossibility result is deliberately stronger than "we have too little data." If safe and unsafe worlds induce exactly the same observations, any randomized protocol accepts with the same probability in both. Uniform false-acceptance control in the unsafe world therefore limits acceptance in the safe world to at most δ. More optimization or a different confidence interval cannot remove this identification boundary; the remedy is new support or an explicit structural assumption.

5 Experimental Protocol

An initial five-seed pilot exposed that maximum class recall gives a constant attacker leakage one; those results are retained as an estimand audit and are excluded from confirmation. The balanced study was then fixed in a committed, hashed preregistration before any seed 5–12 run. It crosses five erasers (INLP, R-LACE, LEACE, TaCo, MANCE++), five datasets (Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, Bios, GaitPDB), and eight untouched seeds, totaling 200 official-method runs. The modalities span natural images, medical images, text, and time series. Bios and CivilComments audit known representation-bias settings (De-Arteaga et al. 2019; Borkan et al. 2019); Camelyon17 is a reliability benchmark, not clinical evidence. Each run records split hashes, train-only preprocessing, its upstream commit and entry point, per-example audit-array hashes, and the shared runner commit. Model cards, datasheets, auditing, and reproducibility guidance motivate this receipt design (Mitchell et al. 2019; Gebru et al. 2021; Raji et al. 2020; Pineau et al. 2021).

The official frontier contains 12 edits per dataset–seed block: INLP ranks $\{1, 2, 4, 8\}$, R-LACE ranks $\{1, 4\}$, closed-

Dataset	Modality	Target Y	Erased/audited source S	Target environment/support
Waterbirds	vision	bird type	background/place	place; supported
Camelyon17-WILDS	medical vision	tumor status	center 0 vs. centers 1–4	hospital; center 2 unseen
CivilComments-WILDS	text	toxicity	identity mention	identity mention; supported
Bios	text	profession (28-way)	gender label	gender label; supported
GaitPDB	time series	Parkinson status	recording site	recording site; supported

Table 1: Cross-modal protocol and support scope. "Supported" means every registered external environment is represented during certification; it does not imply that the bounded-reweighting assumption is known to hold. Camelyon17 is included to instantiate the unsupported-environment boundary, not to make a clinical claim.

form LEACE, TaCo removal counts $\{1, 2, 3, 5\}$, and one registered MANCE++ configuration. Every method calls a pinned upstream entry point; no proxy implementation contributes a row. We cap eraser training, construction, certification, and external evaluation at 8,000, 2,000, 8,000, and 8,000 examples. Train-only standardization and, where needed, randomized PCA precede editing. Eraser fitting, post-edit target-probe fitting, and attacker fitting use no certification outcomes.

Each candidate retrains a logistic target probe and four leakage attackers: logistic regression, an RBF Nystroem map followed by logistic regression, a random forest, and a two-layer MLP. Certification uses fractions $\{0.05, 0.10, 0.25, 0.50, 1.00\}$, target thresholds $\tau \in \{0.05, 0.10, 0.20\}$, leakage thresholds $\lambda \in \{0.60, 0.70, 0.80\}$ and $\delta = 0.05$. The primary fixed profile uses $\Gamma = 1.0$ at the full certification size; a fixed $\Gamma = 1.01$ sensitivity and the full simultaneous radius are also reported. We compare always deploy, feasible point selection without intervals, VERA-IUT, the VERA envelope, and an external oracle. External contract violation is reported separately from procedural non-certifiability. Camelyon17 external balanced leakage is NA because the slice has one source class, and VERA must abstain because center 2 is absent during certification.

Within each configuration, the registered selection utility minimizes the largest validation balanced leakage, then the largest environment-conditional paired harm, then a stable candidate key. Point selection applies this order to empirically feasible edits without intervals. VERA-IUT applies it only to edits passing the fixed-profile certificate. The envelope rule additionally requires the simultaneous common radius to reach the profile. Always-deploy uses the same utility order without a feasibility gate; the external oracle is an evaluation ceiling and never supplies a training or selection label.

We report two endpoints separately. A measured external violation requires an estimable external target and balanced-leakage contract and an actually deployed edit that fails either threshold. A procedurally unsupported deployment enters a registered environment or source-label cell absent from certification. The latter is a failure of identification, not evidence that an observable external threshold was exceeded. This separation prevents the support theorem from manufacturing empirical failures by definition.

The exact balanced study contains 2,000 replicates in each of 54 cells: six certification sizes, three δ levels, and three shift budgets. Its finite multinomial/binomial law gives an analytic abstention probability before Monte Carlo is observed. A separately hashed 216-cell grid varies certification size, four eraser-family sizes plus identity, validated-environment count, and δ, with 2,000 repetitions per cell. The real learning-curve overlay is labeled as a full-certification plug-in diagnostic, not independent theorem validation. Threshold configurations share eight seed-level fits, so an exact one-sided seed-blocked sign-flip test governs the point-selection comparison under its paired sign-symmetry null, with Holm correction over the four externally estimable datasets. A sign-only test is a lower-power sensitivity that does not require magnitude symmetry.

6 Results

Locked headline. VERA gives finite-sample false-acceptance control over its declared shift class and identifies when certification is impossible; on Camelyon17 it forced abstention in all 144 registered VERA configurations because the deployment hospital was outside certification support. This family was selected from pilot seeds 0–4 and locked before all reported seeds 5–12; it is not the average over a searched threshold grid.

Error control and retention. Across all externally estimable primary configurations, VERA-IUT produced 0 measured violations in 288 configurations (0.0%); at the fixed $\Gamma = 1.01$ sensitivity the corresponding result was 0/288 (0.0%). It retained 45/146 externally safe opportunities (30.8%; seed-cluster bootstrap 95% interval 27.5%–34.5%). The preregistered configuration-level Clopper–Pearson interval is retained in the supplement as a dependent-cell diagnostic. Camelyon17 returned forced abstention because center 2 and one required source-class cell are outside certification support.

Paired analysis. Holm-adjusted one-sided seed-blocked sign-flip values for point selection minus VERA-IUT were Bios 1.0000, CivilComments 0.5000, GaitPDB 1.0000, Waterbirds 0.5000. Each seed is one randomization block over all nine correlated thresholds; a sign-only sensitivity and dependent-cell McNemar diagnostics are reported in the supplement.

Exact calibration. The locked synthetic study had 0 false acceptances in 108,000 trials. All 54 observed abstention rates fell inside their simultaneous exact prediction bands, and a separately implemented replay reproduced every cell.

Rule	Deploy	Viol./estimable	Conditional
Always deploy	100.0%	166/288 (57.6%)	57.6%
Point selection	45.8%	14/288 (4.9%)	9.9%
VERA-IUT	12.5%	0/288 (0.0%)	0.0%
VERA envelope	10.0%	0/288 (0.0%)	0.0%
External oracle	40.6%	0/288 (0.0%)	0.0%

Table 2: Primary full-certification results over all nine locked threshold pairs and eight untouched seeds. Violation rates include abstentions in the denominator; the conditional column is descriptive. Camelyon17 external balanced leakage is non-estimable and excluded from measured-violation denominators.

A second locked grid varied candidate count over $M \in \{5, 9, 13, 17\}$ and validated-group count over $|\mathcal{G}| \in \{1, 3, 5\}$; all 216 cells passed simultaneous coverage.

Prespecified stress family. Across 32 pilot-selected, preregistered stress configurations, point selection deployed externally violating edits in 7/32 cases (21.9%), while VERA-IUT deployed none. Waterbirds contributed 3/8 and CivilComments 2/8 at their locked stress thresholds; GaitPDB contributed 1/8, and Bios 0/8. The effect direction is consistent—there were no VERA-only violations—but only three seed blocks were nonzero for Waterbirds and CivilComments. Their one-sided seed-blocked raw values were therefore $p = 0.125$ and Holm-adjusted values were $p = 0.5$. We report the effect size because it is prespecified and receipted, but do not call the paired comparison statistically significant.

What the certificate costs. Over all nine primary threshold pairs, point selection deployed in 45.8% of configurations and violated 14/288 externally estimable contracts. VERA-IUT deployed in 12.5%, violated 0/288, and retained 45/146 external-oracle opportunities. This is a certification tax rather than an accuracy claim: VERA sacrifices deployments when evidence is insufficient. The more conservative simultaneous envelope deployed in 10.0% of configurations. At $\Gamma = 1.01$, IUT deployment fell to 10.3% with 0/288 measured violations; no edit certified at $\Gamma \geq 1.25$ under the locked thresholds.

The attacker portfolio matters. The full four-attacker audit deployed in 12.5% of configurations with zero measured violations. A linear-only audit deployed in 43.6% but violated 98/288 (34.0%) external contracts when judged by the complete portfolio; linear plus RBF still violated 17.7%. Dropping only the MLP doubled deployment to 25.0% and produced 43/288 violations. Forest, MLP, RBF, and linear attackers were the limiting component in 22, 10, 9, and 4 of the 45 full-portfolio deployments, respectively. Thus heterogeneous attackers are not decorative multiplicity: they materially change which edits appear safe.

7 Reproducibility

Code, hashed preregistrations, compact receipts, audit scripts, and manuscript sources are available at https://github.com/RudraChopra/vera-edit-or-abstain. Large public-dataset derivatives remain off GitHub; manifests and content hashes identify them.Every claim-grade result is regenerated from a receipt that records the locked parent commit, upstream eraser commit, split hashes, and per-example audit-array hashes.

The fail-closed receipt audit validates all 200 dataset–method–seed runs and zero proxy rows. A second implementation rehashes 480 raw NPZ archives, reconstructs all 25,920 candidate rows, and replays all 10,800 rule decisions. The first replay exposed a display-name mismatch between upstream 'R-LACE' and registered 'RLACE'; both the failed audit and the label-only correction are retained. After canonicalization, numeric, eligibility, radius, selection, and external-semantic mismatches are all zero.

Generative-AI assistance disclosure. OpenAI Codex was used extensively for research ideation, literature discovery, theorem and proof drafting, software implementation, experiment orchestration, statistical-analysis code, figure and manuscript drafting, and editing. No AI system is an author or a citable source. The human author or authors are responsible for independently verifying every claim, proof, implementation, reference, data right, and policy obligation before submission; that verification remains an explicit presubmission gate.

8 Limitations and Conclusion

VERA's guarantee is finite-sample but scoped. It assumes bounded audit variables, a certification fold independent of edit and attacker construction, and deployment support covered by certification. The density-ratio budget is a declared sensitivity parameter, not an empirically discovered fact. A finite attacker portfolio cannot prove that no measurable algorithm recovers the concept, and suppressing registered leakage does not establish fairness for unmeasured concepts. The Camelyon17 experiment is not clinical validation. Finally, eight algorithmic seeds quantify fitting and split variation but do not make the five benchmark datasets a random sample of deployment domains. The primary sign-flip analysis relies on paired sign symmetry; a sign-only sensitivity is also reported. Both were fixed before those seeds were run, and all four adjusted values are reported regardless of outcome. The observed point-versus-VERA differences did not survive Holm correction, so the empirical comparison is evidence of a practically relevant failure regime, not a definitive population-level superiority claim. Neither internal audits, exact synthetic coverage, nor a polished artifact predicts conference acceptance.

Within those boundaries, VERA changes the output of representation erasure from "apply this edit" to a falsifiable statement: this edit meets these paired target and attacker contracts up to this supported reweighting budget. When the data cannot justify that sentence, abstention is the result rather than a hidden limitation.

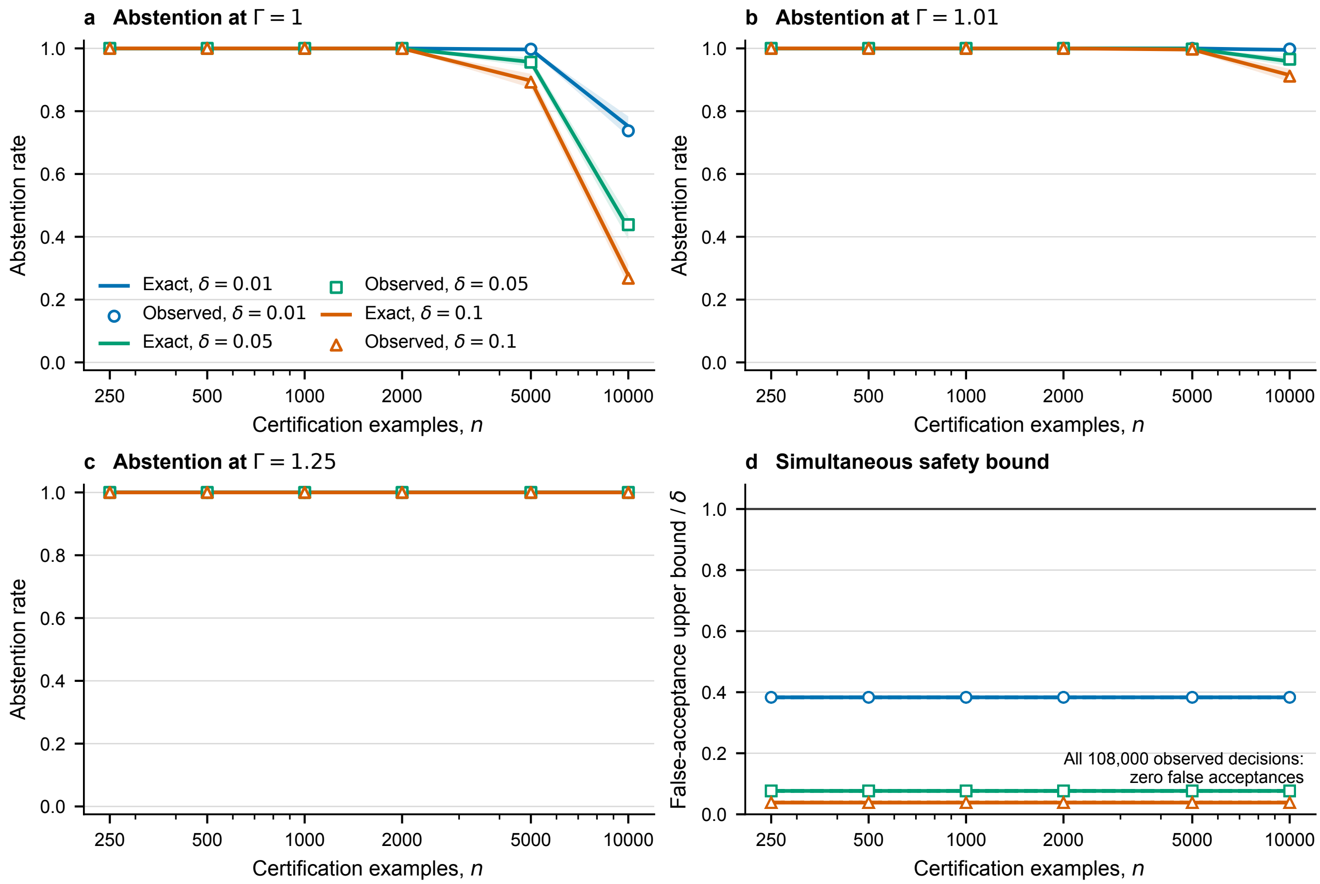

Figure 2: **Exact theory–data match.** Observed abstention from 2,000 repetitions in each of 54 cells overlays the exact prediction; bands are simultaneous over all registered certification sizes. The independent replay verifies every random draw and false-acceptance bound.

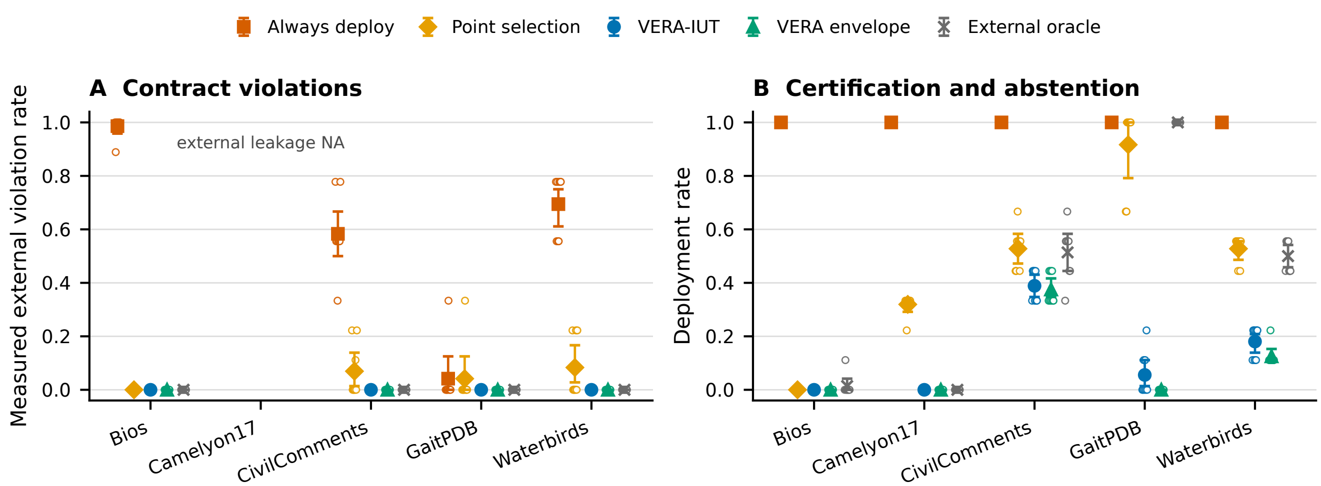

Figure 3: **Deployment rules head to head.** Points are seed-cluster means and intervals resample seed clusters, preserving correlated thresholds and nested fractions within each cluster. Camelyon17 support mismatch is procedural non-certifiability, reported separately from measured external violations.

References

Ai, J.; and Ren, Z. 2024. Not All Distributional Shifts Are Equal: Fine-Grained Robust Conformal Inference. In *International Conference on Machine Learning*.

Akbari, R.; Afshari, M.; and Boddeti, V. 2025. Obliviator Reveals the Cost of Nonlinear Guardedness in Concept Erasure. *Advances in Neural Information Processing Systems*.

Almeida, D. C.; Bravo, J.; Bono, J.; Bizarro, P.; and Figueiredo, M. A. T. 2025. High Probability Risk Control Under Covariate Shift. In *Fourteenth Symposium on Conformal and Probabilistic Prediction with Applications (COPA 2025)*.

Angelopoulos, A. N.; Bates, S.; Candes, E. J.; Jordan, M. I.; and Lei, L. 2025. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. *The Annals of Applied Statistics*.

Angelopoulos, A. N.; Bates, S.; Fisch, A.; Lei, L.; and Schuster, T. 2024. Conformal Risk Control. In *International Conference on Learning Representations*.

Avitan, M.; Goldberg, Y.; and Elazar, Y. 2026. MANCE: Manifold Aware Concept Erasure. arXiv:2607.03973.

Basu Roy Chowdhury, S.; Monath, N.; Dubey, K. A.; Ahmed, A.; and Chaturvedi, S. 2023. Robust Concept Erasure via Kernelized Rate-Distortion Maximization. *Advances in Neural Information Processing Systems*.

Bates, S.; Angelopoulos, A.; Lei, L.; Malik, J.; and Jordan, M. 2021. Distribution-Free, Risk-Controlling Prediction Sets. *Journal of the ACM*.

Belrose, N.; Schneider-Joseph, D.; Ravfogel, S.; Cotterell, R.; Raff, E.; and Biderman, S. 2023. LEACE: Perfect Linear Concept Erasure in Closed Form. *Advances in Neural Information Processing Systems*.

Borkan, D.; Dixon, L.; Sorensen, J.; Thain, N.; and Vasserman, L. 2019. Nuanced Metrics for Measuring Unintended Bias with Real Data for Text Classification. In *Companion Proceedings of The 2019 World Wide Web Conference*.

Brodersen, K. H.; Ong, C. S.; Stephan, K. E.; and Buhmann, J. M. 2010. The Balanced Accuracy and Its Posterior Distribution. In *2010 20th International Conference on Pattern Recognition*.

Cauchois, M.; Gupta, S.; Ali, A.; and Duchi, J. C. 2024. Robust Validation: Confident Predictions Even When Distributions Shift. *Journal of the American Statistical Association*.

Chowdhury, S. B. R.; Dubey, K. A.; Beirami, A.; Kidambi, R.; Monath, N.; Ahmed, A.; and Chaturvedi, S. 2025. Fundamental Limits of Perfect Concept Erasure. In *International Conference on Artificial Intelligence and Statistics*.

David, S. B.; Lu, T.; Luu, T.; and Pal, D. 2010. Impossibility Theorems for Domain Adaptation. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*.

De-Arteaga, M.; Romanov, A.; Wallach, H.; Chayes, J.; Borgs, C.; Chouldechova, A.; Geyik, S.; Kenthapadi, K.; and Kalai, A. T. 2019. Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*.

Deka, N.; and Sutherland, D. J. 2023. MMD-B-Fair: Learning Fair Representations with Statistical Testing. In *International Conference on Artificial Intelligence and Statistics*.

Ehyaei, A.-R.; Farnadi, G.; and Samadi, S. 2025. From Fragile to Certified: Wasserstein Audits of Group Fairness Under Distribution Shift. arXiv:2509.26241.

El-Yaniv, R.; and Wiener, Y. 2010. On the Foundations of Noise-free Selective Classification. *Journal of Machine Learning Research*.

Elazar, Y.; Ravfogel, S.; Jacovi, A.; and Goldberg, Y. 2021. Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals. *Transactions of the Association for Computational Linguistics*.

Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J. W.; Wallach, H.; III, H. D.; and Crawford, K. 2021. Datasheets for Datasets. *Communications of the ACM*.

Geifman, Y.; and El-Yaniv, R. 2017. Selective Classification for Deep Neural Networks. *Advances in Neural Information Processing Systems*.

Geifman, Y.; and El-Yaniv, R. 2019. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. In *International Conference on Machine Learning*.

Hewitt, J.; and Liang, P. 2019. Designing and Interpreting Probes with Control Tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*.

Holstege, F.; Ravfogel, S.; and Wouters, B. 2025. Preserving Task-Relevant Information Under Linear Concept Removal. *Advances in Neural Information Processing Systems*.

Huang, J.; Zeng, H.; Jing, B.; Wei, H.; and An, B. 2026. Conditional Performance Guarantee for Large Reasoning Models. arXiv:2601.22790.

Jeong, S.; and Namkoong, H. 2020. Robust Causal Inference under Covariate Shift via Worst-Case Subpopulation Treatment Effects. In *Conference on Learning Theory*.

Jourdan, F.; Bethune, L.; Picard, A.; Risser, L.; and Asher, N. 2023. TaCo: Targeted Concept Erasure Prevents Non-Linear Classifiers From Detecting Protected Attributes. arXiv:2312.06499.

Jovanovic, N.; Balunovic, M.; Dimitrov, D. I.; and Vechev, M. 2023. FARE: Provably Fair Representation Learning with Practical Certificates. In *International Conference on Machine Learning*.

Kang, M.; Li, L.; Weber, M.; Liu, Y.; Zhang, C.; and Li, B. 2022. Certifying Some Distributional Fairness with Subpopulation Decomposition. *Advances in Neural Information Processing Systems*.

Koh, P. W.; Sagawa, S.; Marklund, H.; Xie, S. M.; Zhang, M.; Balsubramani, A.; Hu, W.; Yasunaga, M.; Phillips, R. L.; Gao, I.; Lee, T.; David, E.; Stavness, I.; Guo, W.; Earnshaw, B.; Haque, I.; Beery, S. M.; Leskovec, J.; Kundaje, A.; Pierson, E.; Levine, S.; Finn, C.; and Liang, P. 2021. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *International Conference on Machine Learning*.

Laufer-Goldshtein, B.; Fisch, A.; Barzilay, R.; and Jaakkola, T. 2023. Efficiently Controlling Multiple Risks with Pareto Testing. In *International Conference on Learning Representations*.

Mitchell, M.; Wu, S.; Zaldivar, A.; Barnes, P.; Vasserman, L.; Hutchinson, B.; Spitzer, E.; Raji, I. D.; and Gebru, T. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*.

Najafi, A.; Sani, S. M.; and Farnia, F. 2025. Certifiably Robust Model Evaluation in Federated Learning under Meta-Distributional Shifts. In *International Conference on Machine Learning*.

Pimentel, T.; Valvoda, J.; Maudslay, R. H.; Zmigrod, R.; Williams, A.; and Cotterell, R. 2020. Information-Theoretic Probing for Linguistic Structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Pineau, J.; Vincent-Lamarre, P.; Sinha, K.; Lariviere, V.; Beygelzimer, A.; d'Alche Buc, F.; Fox, E.; and Larochelle, H. 2021. Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program). *Journal of Machine Learning Research*.

Raji, I. D.; Smart, A.; White, R. N.; Mitchell, M.; Gebru, T.; Hutchinson, B.; Smith-Loud, J.; Theron, D.; and Barnes, P. 2020. Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*.

Ravfogel, S.; Elazar, Y.; Gonen, H.; Twiton, M.; and Goldberg, Y. 2020. Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Ravfogel, S.; Twiton, M.; Goldberg, Y.; and Cotterell, R. D. 2022a. Linear Adversarial Concept Erasure. In *International Conference on Machine Learning*.

Ravfogel, S.; Vargas, F.; Goldberg, Y.; and Cotterell, R. 2022b. Adversarial Concept Erasure in Kernel Space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.

Sagawa, S.; Koh, P. W.; Hashimoto, T. B.; and Liang, P. 2020. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. In *International Conference on Learning Representations*.

Shao, S.; Ziser, Y.; and Cohen, S. B. 2023. Erasure of Unaligned Attributes from Neural Representations. *Transactions of the Association for Computational Linguistics*.

Sohoni, N.; Dunnmon, J.; Angus, G.; Gu, A.; and Re, C. 2020. No Subclass Left Behind: Fine-Grained Robustness in Coarse-Grained Classification Problems. *Advances in Neural Information Processing Systems*.

Subbaswamy, A.; Adams, R.; and Saria, S. 2021. Evaluating Model Robustness and Stability to Dataset Shift. In *International Conference on Artificial Intelligence and Statistics*.

Thomas, P.; and Learned-Miller, E. 2019. Concentration Inequalities for Conditional Value at Risk. In *International Conference on Machine Learning*.

Tibshirani, R. J.; Foygel Barber, R.; Candes, E.; and Ramdas, A. 2019. Conformal Prediction Under Covariate Shift. *Advances in Neural Information Processing Systems*.

Timans, A.; Verma, R.; Nalisnick, E.; and Naesseth, C. A. 2025. On Continuous Monitoring of Risk Violations under Unknown Shift. In *Conference on Uncertainty in Artificial Intelligence*.

Voita, E.; and Titov, I. 2020. Information-Theoretic Probing with Minimum Description Length. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Waudby-Smith, I.; and Ramdas, A. 2024. Estimating Means of Bounded Random Variables by Betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*.

Yu, X.; and Liu, J. 2026. A Joint Finite-Sample Certificate for Adaptive Selective Conformal Risk Control. arXiv:2606.08517.

Zollo, T.; Morrill, T.; Deng, Z.; Snell, J.; Pitassi, T.; and Zemel, R. 2024. Prompt Risk Control: A Rigorous Framework for Responsible Deployment of Large Language Models. In *International Conference on Learning Representations*.